

MA7444 Excel for Data Science

Airbnb Listings Business Report

Introduction and Objectives:

The present report is an analysis of 12,805 listings on Airbnb from different countries, locations, property types and prices. The analysis aimed to identify concentrations of listings, compare pricing patterns, and highlight factors associated with unusually high or low prices.

The analysis was performed in Microsoft Excel using data-cleaning formulas, validation checks, summary statistics, PivotTables, PivotCharts and interactive slicers. A dashboard was created to present the key findings and enable users to filter the results by country, property type and price band.

It also looks at the distribution of listings by check-in period. However, this is used only as a time-related comparison, since the dataset does not contain historical listing dates or availability records suitable for a true time series evaluation.

Dataset Overview and Data Preparation

Dataset Overview: The dataset consisted of 12,805 records of Airbnb listings and 23 original variables. The available fields were listing and host identifiers, listing name, address, country, price, rating, number of reviews, guest capacity, bedrooms, beds, bathrooms, amenities, house rules, check-in time, checkout time.

The cleaned dataset contained listings from 119 countries. The descriptive price results were:

Measure

	Result
No of listing	12,805
Minimum price	393
First quartile	3,994
Median Price	8,175
Third quartile	16,062
Average price	17,697.80

Maximum price	1,907,963
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The mean price was significantly above the median suggesting a positive skew in the distribution. The difference was driven by a small number of listings with exceptionally high prices.

Price data were analysed as provided in the dataset. No common currency was specified so no currency symbols were used and direct international price comparisons should be viewed with caution.

Data Preparation:

The original imported dataset was saved in a separate Raw_Data sheet. A duplicate copy in Clean_Data was cleaned and derived calculations were performed. This maintained the original data and provided a clear audit trail.

The country names were cleaned using TRIM function for removing any unnecessary leading/trailing spaces. The location was extracted from the 1st part of the address by applying text functions. This yielded a usable location variable, though not always a formally verified neighbourhood.

Listing IDs were imported and saved as text. This was necessary because some of the IDs had more than 15 digits (the precision limit of Excel for numbers). Initially, by saving the IDs as numbers, we got 13 false duplicates because of the rounding of the last digits in Excel. The conversion of the identifiers to text also led to a count of zero for duplicate-ID.

Those listed as “New” were split out with numeric ratings. Numerical values are converted with VALUE and IFERROR . New listings are categorized separately. Review counts were also converted to numeric values by removing the comma separator.

Additional variables were created to aid in the analysis:

country_clean (standardized country names), location_clean (first part of address)
rating_numeric and rating_status, reviews_numeric, price_per_guest, property_type
price_band, checkin_period, price_outlier, record_check.

The property type was not provided in the original dataset. We used keywords in the listing names to estimate it, including such terms as villa, apartment, house, studio, cottage, hotel and room. Any listing that did not have a clear keyword was put into the category “Other.”

Validation and Data-Quality Checks:

Validation checks are created in separate data_checks worksheet. results are:

Validation check	Result
Clean records	12,805
Duplicate IDs	0
Missing IDs	0
Missing listing names	0
Missing cleaned countries	0
Missing prices	0
Zero or negative prices	0
Invalid guest capacities	0
New or unrated listings	4,238
Listing with numeric rating	8,567
Missing host names	8
Missing check in values	800
Missing checkout values	2,450

Missing check-in, checkout and host-name values were kept, since there was still sufficient information in the records for the other parts of the analysis.

Outliers in price were identified using the interquartile-range method:

$$\text{Upper boundary} = Q3 + 1.5 * \text{IQR}$$

The upper boundary calculated was 34,164. 1,273 listings were identified as high-price outliers. These records comprised approximately 9.9% of the dataset. They were flagged, not deleted, so that unusual pricing patterns could be observed and communicated on the dashboard.

These preparation and validation procedures supported the coursework requirement to document the methods calculations and checks used in the analysis

Analysis and Key Findings

Country Pricing Patterns

India had the highest number of listings (2,779 records). But its average price was around 7,536.81, which was significantly less than a few countries with less listings.

The most represented countries, Turkey, Italy and Greece, posted much higher average prices:

Row Labels	Listing Count	Average of price_per_guest	Average Price	Average of rating_numeric
India	2779	£1,536.86	£7,536.81	4.823812117
Italy	1187	£4,544.99	£31,635.98	4.878543956
Greece	889	£5,750.12	£31,431.55	4.871356383
Thailand	683	£3,176.82	£21,021.27	4.844056122
Turkey	522	£6,075.19	£32,863.13	4.822745098
France	466	£3,762.06	£18,826.93	4.846610169
Morocco	377	£1,882.79	£9,370.47	4.830211864
Georgia	371	£1,660.96	£6,796.86	4.860923695
Japan	322	£1,833.70	£10,416.48	4.854225806
United Kingdom	313	£5,408.87	£13,914.43	4.936969697
Grand Total	7909	£3,221.81	£17,763.11	4.852825743

The results show that listing volume and average price were not directly related. India had the largest supply but relatively low average prices, compared with Turkey, Italy and Greece which had lower listing volumes but much higher average prices.

The results should be interpreted with caution as there was no standardised currency field in the data set. Hence, differences between countries may be a reflection of market conditions, as well as differences in currencies used in the original source.



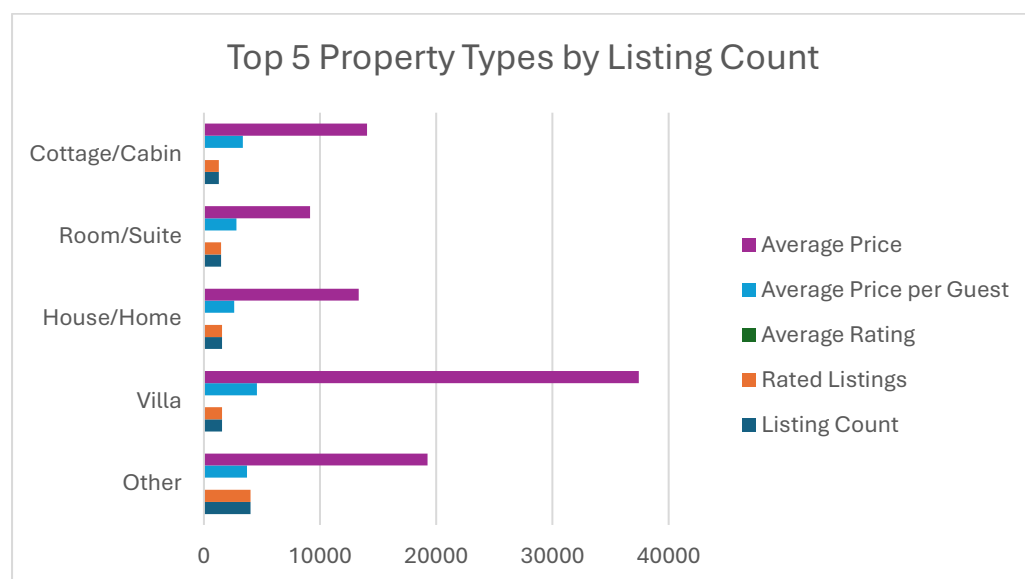
Property Type Patterns

Row Labels	Listing Count	Rated Listings	Average Rating	Average Price per Guest	Average Price
Other	4025	4025	4.86	£3,700.85	£19,252.41
Villa	1556	1556	4.86	£4,559.08	£37,441.17
House/Home	1553	1553	4.86	£2,603.45	£13,312.00
Room/Suite	1482	1482	4.84	£2,811.11	£9,146.68
Cottage/Cabin	1282	1282	4.87	£3,339.28	£14,039.40
Grand Total	9898	9898	4.86	£3,483.54	£18,991.40

Villas had the highest average price among the five biggest categories, at around 37,441.17. This price was more than four times the average price for Room/Suite listings.

The largest category was Other, with the most listings. This suggests many listing names did not contain a clear keyword to reliably assign a property type. The category should therefore be viewed as a limitation of the classification methodology rather than as an actual property type.

The average ratings of the largest property categories were relatively similar, generally between about 4.84 and 4.87. Thus, the price differences between the categories were greater than the guest rating differences.



Location Concentration

Row Labels	Listing Count	Rated Listings	Average Price	Average Price per Guest	Average of rating_numeric
Pantelleria	203	203	£17,760.98	3943.15	4.88
New Delhi	194	194	£7,694.91	2036.42	4.82
Gudauri	191	191	£6,980.69	1628.39	4.86
Manali	182	182	£5,919.02	1173.61	4.85
Marrakesh	114	114	£12,841.79	2428.51	4.85
Trinidad	105	105	£2,395.30	654.99	4.85
Sapanca	99	99	£19,544.16	4024.06	4.84
Marrakech	82	82	£14,172.21	2659.60	4.83
Rishikesh	70	70	£4,576.97	1359.33	4.82
Srinagar	68	68	£4,420.53	1356.12	4.82
Grand Total	1308	1308	£9,894.73	2193.89	4.85

The derived locations with the highest listing count was Pantelleria however the difference between the four largest locations was relatively small.

Average prices were all over the place. Average price in Pantelleria was about 17,903.34 and average price in Manali was about 5,919.02. The results suggest that geographical location may be an important contributor to price variation.

In addition to inconsistent place naming, the analysis identified . For example, Marrakesh and Marrakech were two different values. These names may refer to the same geographical area, but they were not merged since the dataset did not contain verified geographical identifiers.

Check in Period Analysis

Row Labels	Count of id	Average of price
Afternoon	8063	20688.40233
Evening	346	19061.03757
Midday	2002	10993.55744
Morning	668	14019.73802
Grand Total	11079	18483.61711

Afternoon check-in was the biggest category, with more than 8,000 listings. It also had the highest average price at around 20,688.40.

The midday listings had the lowest average price, at around 10,993.56. But the check-in period cannot be seen as the direct reason for the price differences. Or perhaps higher-priced property types or locations are just more likely to have afternoon or evening check-in.

This is a time-related comparison rather than a historical time-series evaluation as the dataset did not contain listing dates, booking dates or repeated availability observations.

Dashboard Overview

An interactive Excel dashboard was developed to summarise the main findings of the analysis. The dashboard contains key performance indicators, comparative charts and slicers so that users can explore pricing and listing patterns without looking at the underlying worksheets.

As for the main indicators shown on the dashboard, they are:

The dashboard also provides visual comparison for:

average price for the countries with highest volume of listings;

top 5 derived property types by number of listings;

listing distribution or average price by check-in period;

listing distribution by price bands.

Interactive slicers were supplied for Country, Property Type & Price Band. These controls allow the user to focus on selected market segments and at the same time update the PivotTables, charts and headline indicators.

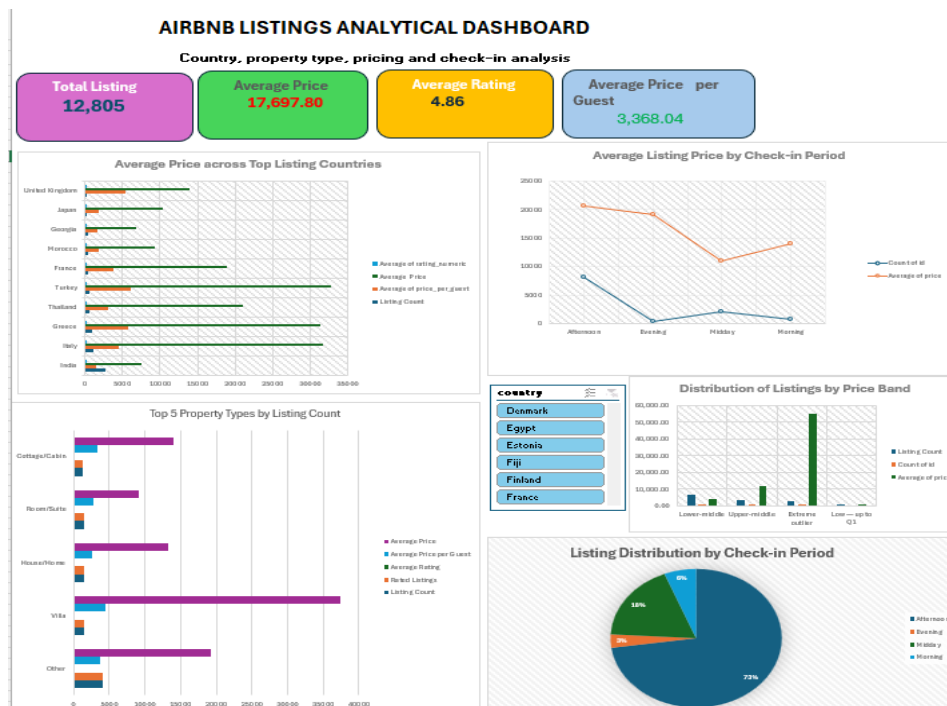
The property-type chart shows that villas are associated with substantially higher average prices, while Rooms/Suites are generally positioned at the lower end of the price range. The large Other category also demonstrates the limitation of deriving property types from listing-name keywords.

The price-band chart communicates the uneven price distribution. Although extreme outliers represent fewer than one in ten listings, their very high prices have a considerable effect on the overall average.

The dashboard therefore provides both a high-level summary and an interactive method of examining individual segments. It satisfies the coursework requirement for selected-country pricing summaries, Top 5 comparisons, time-related analysis, unusual-price highlights and interactive controls.

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Dataset Limitations

First, the dataset does not contain a standardised currency variable. Prices from different countries might thus be in different currencies. Direct comparisons of national averages may then not reflect real differences in affordability or market value.

Second, property type was not considered as an independent variable. It was based on keywords in the listing names. Listings that had unclear names were grouped as Other, and some listings may have been incorrectly classified.

Third, the location was extracted from the first component of the address. This was used as a proxy for location or neighbourhood measure but was not supported by verified geographical identifiers. Different spellings, such as Marrakesh and Marrakech, can divide listings from the same area.

Finally, there are large outliers in price in the dataset. These values could be for luxury properties, for different currencies, data entry errors, or for prices in different units. They were left in as there was not enough evidence to call them errors. Their presence increases the mean, but makes the median a more reliable measure of a typical listing price.

Business Implications

The analysis gives several practical insights for Airbnb hosts, market analysts and accommodation businesses.

The large variation in average price across countries and locations suggests that pricing decisions should be based on local market conditions, rather than on the average of the whole data set. The overall average is heavily skewed by unusually high priced listings and may therefore not be a good benchmark. Hosts should take a look at their own listings and similar properties in the same area, and use the median or relevant price band as a more realistic reference point.

The price variations also seem to be very heavily influenced by the type of property. Average price of Villas was significantly higher than that of Houses/Homes and Rooms/Suites. This may reflect differences in capacity, privacy, facilities and target customer. This means that hosts with premium features may be able to command higher prices if they are clearly highlighted in the listing.

Affordability, guest ratings, location and value per guest may allow lower-priced categories and Rooms/Suites to compete more effectively. Price per guest is especially useful when comparing properties of different capacities, as the total price of the listing can make larger properties appear disproportionately expensive.

The high-price outliers should be treated apart from the normal listings. These properties may be luxury accommodation, very large properties or inconsistent price units. Including them in a general pricing benchmark, you may be making unrealistic pricing decisions.

The dashboard enables decision-making by allowing users to filter results by country, property type and price band. This allows comparison of a selected market segment with the total data set and to find differences in listing volume, average price and customer ratings.

Conclusion :

In this project, Microsoft Excel was used to prepare, validate, analyze and visualize a dataset of 12,805 Airbnb listings. The analysis looked at differences by country, derived location, property type, price band and check-in period.

The results showed that listing volume was not necessarily linked to higher average prices. India had the highest number of listings, but a lower average price than some countries represented far less often. Property type also showed significant price differences, villas being the most expensive of the major categories on average.

The price distribution was very positively skewed. The mean price of 17,697.80 was much higher than the median of 8,175, mainly because of 1,273 high-price outliers. Thus, the median and segmented price bands provided more representative measures of typical listing prices.

Afternoon had the highest average price and was the most popular check-in period of the main time categories. However, this relationship should not be seen as a causal one, and the lack of historical dates did not allow a real time-series evaluation.

The interactive dashboard summarized the key findings through KPI indicators, charts and slicers. This provided a neat way to drill down into country, property type and price band trends.

The report therefore fulfils the requirement of summarising the main findings and reflecting on the limitations of the dataset.